

P1. a) Let the quadratic spline S_2 be

$$S_2(x) = a_j + b_j(x - x_j) + c_j(x - x_j)^2 \quad \text{for } x \in [x_j, x_{j+1}]$$

$j = 0, 1, \dots, N-1$

let $h_j = x_{j+1} - x_j$, $P_j = S_2|_{[x_j, x_{j+1}]}$, then we have.

$$\begin{cases} P_j(x_j) = f_j & - \textcircled{1} \\ P_j(x_{j+1}) = f_{j+1} & - \textcircled{2} \quad \text{for } j = 0, 1, \dots, N-1 \\ P_j'(x_{j+1}) = P_{j+1}'(x_{j+1}) & - \textcircled{3} \quad \text{for } j = 0, 1, \dots, N-2. \end{cases}$$

$$\textcircled{1} \Rightarrow a_j = f_j \quad - \textcircled{1}'$$

$$\textcircled{2} \Rightarrow a_j + b_j h_j + c_j h_j^2 = f_{j+1} \quad - \textcircled{2}' \quad j = 0, 1, \dots, N-1$$

$$\textcircled{3} \Rightarrow b_j + 2c_j h_j = b_{j+1} \Rightarrow c_j = \frac{b_{j+1} - b_j}{2h_j} \rightarrow \textcircled{3}' \quad j = 0, 1, \dots, N-2$$

put $\textcircled{3}'$ & $\textcircled{1}'$ in $\textcircled{2}'$, we get

$$f_j + b_j h_j + \frac{b_{j+1} - b_j}{2h_j} \cdot h_j^2 = f_{j+1}$$

$$(b_j + b_{j+1}) = \frac{2}{h_j} (f_{j+1} - f_j) \quad - \textcircled{4}' \quad j = 0, \dots, N-2$$

The additional condition $S_2'(x_0) = f_0'$ means $b_0 = f_0'$

so we have the following algorithm

$$b_0 = f_0', a_0 = f_0,$$

For $j = 0, \dots, N-2$

$$b_{j+1} = \frac{2}{h_j} (f_{j+1} - f_j) - b_j$$

$$c_j = \frac{b_{j+1} - b_j}{2h_j}$$

$$a_{j+1} = f_{j+1}$$

end.

$$c_{N-1} = \frac{f_N - a_{N-1} - b_{N-1} h_{N-1}}{h_{N-1}^2}$$

(b). The periodic boundary condition $S_2'(x_0) = S_2'(x_N)$ means.

$$b_0 = b_N$$

We have the following system with (4)'

$$(*) \quad \begin{bmatrix} 1 & 1 & 0 & \dots & 0 \\ 0 & 1 & 1 & \dots & 0 \\ 0 & 0 & 1 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & 0 & 0 & \dots & 1 \end{bmatrix} \begin{bmatrix} b_0 \\ b_1 \\ \vdots \\ b_{N-1} \end{bmatrix} = \begin{bmatrix} g_0 \\ g_1 \\ \vdots \\ g_{N-1} \end{bmatrix} \quad \text{where } g_j = \frac{2}{h_j} (f_{j+1} - f_j) \quad j=0, 1, \dots, N-1.$$

It is easy to find.

$$\det(A) = 1 + (-1)^{N-1}$$

(I) If N is odd, $\det(A) = 2$. i.e., A is nonsingular, then the system (*) is uniquely solvable.

(II). If N is even, $\det(A) = 0$ - i.e. A is singular, then

$$A = LU$$

$$L = \begin{bmatrix} 1 & & & & \\ & 1 & & & \\ & & 1 & & \\ & & & \ddots & \\ 1 & -1 & 1 & -1 & \dots & 1 \end{bmatrix}_{N \times N}$$

$$U = \begin{bmatrix} 1 & 1 & & & \\ & 1 & 1 & & \\ & & 1 & 1 & \\ & & & \ddots & \ddots \\ 0 & \dots & 0 & 0 & 1 \end{bmatrix}_{\text{full rank.}}$$

$$Ab = g \Rightarrow LUb = g \Rightarrow Ub = L^{-1}g, \quad L^{-1} = \begin{bmatrix} 1 & & & & \\ & 1 & & & \\ & & 1 & & \\ & & & \ddots & \\ -1 & 1 & -1 & 1 & \dots & 1 \end{bmatrix}_{N \times N}.$$

$$\Rightarrow (L^{-1}g)_N = 0 \Rightarrow \sum_{j=0}^{N-1} (-1)^{j-1} \frac{2}{h_j} (f_{j+1} - f_j) = 0 \quad (5)$$

so to make $Ab = g$ solvable, (5) must be satisfied, otherwise, there is no solution, i.e. $S_2(x)$ doesn't exist.

P3. a. Let

$$P(x) = \sum_{k=-\frac{N}{2}}^{\frac{N}{2}} d_k e^{ikx} = \sum_{k=0}^N d_{k-\frac{N}{2}} e^{ikx} e^{-i\frac{N}{2}x} = e^{-i\frac{N}{2}x} \sum_{k=0}^N d_{k-\frac{N}{2}} e^{ikx}$$

We apply the interpolation condition to $P(x)$, i.e.

$$P(x_i) = f_i \quad \text{for } i=0, \dots, N$$

$$\text{We get } e^{i\frac{N}{2}x_i} P(x_i) = e^{i\frac{N}{2}x_i} f_i, \quad i=0, \dots, N$$

$$(*) \quad \underbrace{\begin{bmatrix} 1 & e^{ix_0} & e^{i2x_0} & \dots & e^{iNx_0} \\ 1 & e^{ix_1} & e^{i2x_1} & \dots & e^{iNx_1} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & e^{ix_N} & e^{i2x_N} & \dots & e^{iNx_N} \end{bmatrix}}_A \begin{bmatrix} d_{-\frac{N}{2}} \\ \vdots \\ d_0 \\ \vdots \\ d_{\frac{N}{2}} \end{bmatrix} = \begin{bmatrix} f_0 e^{i\frac{N}{2}x_0} \\ f_1 e^{i\frac{N}{2}x_1} \\ \vdots \\ f_N e^{i\frac{N}{2}x_N} \end{bmatrix}$$

$$\det A = \prod_{0 \leq k < l \leq N} (e^{ix_l} - e^{ix_k})$$

Since $a \leq x_0 < x_1 < \dots < x_N < a+2\pi$, then
 $e^{ix_l} \neq e^{ix_k}$ for $k \neq l$.

$$\Rightarrow \det A \neq 0$$

So there exists a unique $\{d_j\}_{j=-\frac{N}{2}}^{\frac{N}{2}}$ satisfying (*).

Let $d_k = a_k + ib_k$ for $k = -\frac{N}{2}, \dots, \frac{N}{2}$, then

$$\begin{aligned} P(x) &= \sum_{k=-\frac{N}{2}}^{\frac{N}{2}} d_k e^{ikx} \\ &= d_0 + \sum_{k=1}^{\frac{N}{2}} d_k e^{ikx} + \sum_{k=1}^{\frac{N}{2}} d_{-k} e^{-ikx} \\ &= d_0 + \sum_{k=1}^{\frac{N}{2}} [(a_k + ib_k)(\cos kx + i \sin kx) + (a_{-k} + ib_{-k})(\cos kx - i \sin kx)] \\ &= d_0 + \sum_{k=1}^{\frac{N}{2}} [(d_k + d_{-k}) \cos kx + (d_k - d_{-k}) i \sin kx] \end{aligned}$$

Let $A_0 = 2d_0$, $A_k = d_k + d_{-k}$, $B_k = i(d_k - d_{-k})$. Then

$$T(x) = P(x)$$

satisfies the interpolation condition.

(b)

$$P(x) = \sum_{k=-\frac{N}{2}}^{\frac{N}{2}} d_k e^{ikx}$$

$$\overline{P(x)} = \sum_{k=-\frac{N}{2}}^{\frac{N}{2}} \overline{d_k} e^{-ikx}$$

$$P(x_j) = f_j = \overline{f_j} = \overline{P(x_j)} \quad \text{for } j=0, \dots, N.$$

$\Rightarrow Q(x) = P(x) - \overline{P(x)}$ has $(N+1)$ zeros.

$$= \sum_{k=-\frac{N}{2}}^{\frac{N}{2}} (d_k - \overline{d_{-k}}) e^{ikx}$$

$$= e^{-\frac{N}{2}ix} \sum_{k=0}^N (d_{k-\frac{N}{2}} - \overline{d_{\frac{N}{2}+k}}) (e^{ix})^k.$$

$$\Rightarrow Q(x) = 0$$

$$\Rightarrow d_k - \overline{d_{-k}} = 0 \quad \dots \quad d_k = \overline{d_{-k}}$$

$$\Rightarrow B_k = i(d_k - d_{-k}) \in \mathbb{R}$$

$$d_0 = \overline{d_0} \Rightarrow d_0 \in \mathbb{R}.$$

$$A_k = d_k + d_{-k} \in \mathbb{R}.$$

$\Rightarrow$ all A_j 's, B_j 's are real.